GENREation

Music Genre Classification Using Convolutional Neural Networks

1. INTRODUCTION AND PROJECT OVERVIEW

GENREation is an audio-classification application that predicts the musical genre of an uploaded recording. It combines digital audio processing, feature extraction and deep learning in one workflow. A user uploads a music file through a Streamlit interface, the application converts the audio into numerical features, passes those features to a trained Convolutional Neural Network (CNN), and displays the predicted genre and the model's class scores. It can also show a log-Mel spectrogram, a visual representation of how audio energy varies over time and frequency. The project demonstrates an end-to-end applied AI workflow: dataset preparation, preprocessing, model training, validation, evaluation, model saving and interactive prediction.

2. PROBLEM STATEMENT

Music collections contain recordings from many styles and assigning a genre manually can be time-consuming and subjective. Some genres share instruments, rhythms, tempo ranges and production characteristics, which makes classification challenging. GENREation treats this as a supervised multi-class classification problem: *given an audio track, the model estimates which of ten learned genre categories best matches its acoustic patterns. The system is an educational classification tool; its output is not a definitive judgement of musical identity.*

3. OBJECTIVES

The project aims to:

1. Prepare a labelled dataset for supervised music classification.
2. Extract consistent MFCC features from audio recordings.
3. Design and train a CNN to learn useful patterns from those features.
4. Evaluate the model on held-out data using accuracy and class-wise metrics.
5. Save the trained model and genre-label mapping for reuse.
6. Develop an interactive application for audio upload, prediction and visualization.

4. DATASET DESCRIPTION

The project uses the GTZAN Genre Collection, a commonly used educational benchmark containing ten labelled categories: Blues, Classical, Country, Disco, Hip-Hop, Jazz, Metal, Pop, Reggae and Rock. The standard collection contains 100 tracks per genre (1,000 tracks in total), with recordings approximately 30 seconds long. The training pipeline searches for genre folders either directly inside the dataset directory or within a *genres_original* subfolder. Files that cannot be decoded or processed are skipped and should be accounted for in the run logs. GTZAN is a benchmark dataset, not a complete representation of all music styles, artists or recording conditions.

5. AUDIO PREPROCESSING AND MFCC FEATURE EXTRACTION

Audio is loaded with Librosa, converted to mono and resampled to 22,050 Hz so that recordings with different source sampling rates use a consistent input rate. The pipeline standardizes duration to 30 seconds: shorter recordings are padded and longer recordings are truncated. It then extracts 40 Mel-Frequency Cepstral Coefficients (MFCCs), using an FFT size of 2,048 and a hop length of 512 samples. MFCCs summarize the short-term spectral envelope using a perceptually motivated frequency scale and provide a compact representation of timbral information. Each coefficient is normalized across its time frames. With the project's stated settings, the feature tensor has the shape 40 × 1,292 × 1. The same shared function in src/audio_features.py is used for training and prediction, ensuring consistent preprocessing at inference time.

6. DATASET SPLIT AND CNN ARCHITECTURE

The labelled tracks are divided into training and validation subsets using a stratified 80/20 split. Stratification preserves approximate genre proportions in both subsets, and splitting at track level keeps each original track wholly within one subset. The CNN receives the two-dimensional MFCC representation as a single-channel input. It contains three convolutional blocks with 32, 64 and 128 filters. Each block uses a 3 × 3 convolution with ReLU activation, batch normalization and 2 × 2 max pooling. Convolutional filters learn local patterns in the time–coefficient representation; batch normalization stabilizes activations; and max pooling reduces feature-map dimensions. Global average pooling summarizes the feature maps, followed by a 128-unit dense layer and dropout (0.4). The final ten-unit softmax layer produces a score for each genre.

7. MODEL TRAINING AND OPTIMIZATION

The model is trained with the Adam optimizer and sparse categorical cross-entropy loss, suitable for integer-encoded labels in a multi-class task. Accuracy is tracked during training and validation. Early stopping monitors validation loss and restores the best weights when progress stalls. ReduceLROnPlateau lowers the learning rate when validation loss stops improving, while model checkpointing saves the selected model. The saved Keras model must be used with its matching genre-label mapping so that each output index is translated to the correct genre name.

8. APPLICATION DESIGN AND PREDICTION WORKFLOW

The user-facing application is implemented in Streamlit. A user uploads an audio file in a supported format, subject to the audio codecs installed in the runtime. The application validates the input, applies the shared preprocessing function, adds the batch dimension and sends the MFCC tensor to the saved CNN. The model returns ten class scores. The application ranks the scores and displays the highest-scoring genre, its score and the top-three alternatives. It can also display a log-Mel spectrogram. The displayed softmax score is the model's output for a class; without a separate calibration procedure, it must not be interpreted as a calibrated probability or a guarantee of correctness.

9. EVALUATION AND RESULTS

The original baseline evaluation supplied during development used 200 validation tracks. It reported 58.50% validation accuracy, equivalent to 117 correct classifications out of 200. Macro precision was 67.93%, macro recall was 58.50%, and macro F1-score was 54.47%. Macro-averaging gives each genre equal weight and helps reveal uneven class performance. The confusion matrix showed correct predictions of Classical 20/20, Hip-Hop 19/20, Metal 17/20, Country 17/20, Blues 10/20, Jazz 11/20, Pop 11/20, Reggae 6/20, Disco 3/20 and Rock 3/20. The results show that the model learns useful genre-related patterns, but performance varies substantially across classes.

10. LIMITATIONS, FUTURE WORK AND CONCLUSION

Performance is limited by the size and characteristics of the dataset, overlap between musical genres, recording quality, noise, short clips and music unlike the training examples. GTZAN is a widely used educational benchmark but has known data-quality and duplication concerns; a random track-level split does not establish artist-independent generalization. Future work includes duplicate and near-duplicate checks, artist-independent evaluation where reliable metadata exists, segment-based training while keeping segments from the same track in one split, data augmentation, log-Mel spectrogram-based CNN experiments and controlled hyperparameter comparisons. A later experimental model performed substantially worse than the original baseline and is not treated as the selected model. GENREation provides a complete audio-classification workflow and a working prediction interface, while its evaluation identifies clear opportunities to improve balanced genre recognition.

Validation accuracy	Macro precision	Macro recall	Macro F1
58.50%	67.93%	58.50%	54.47%